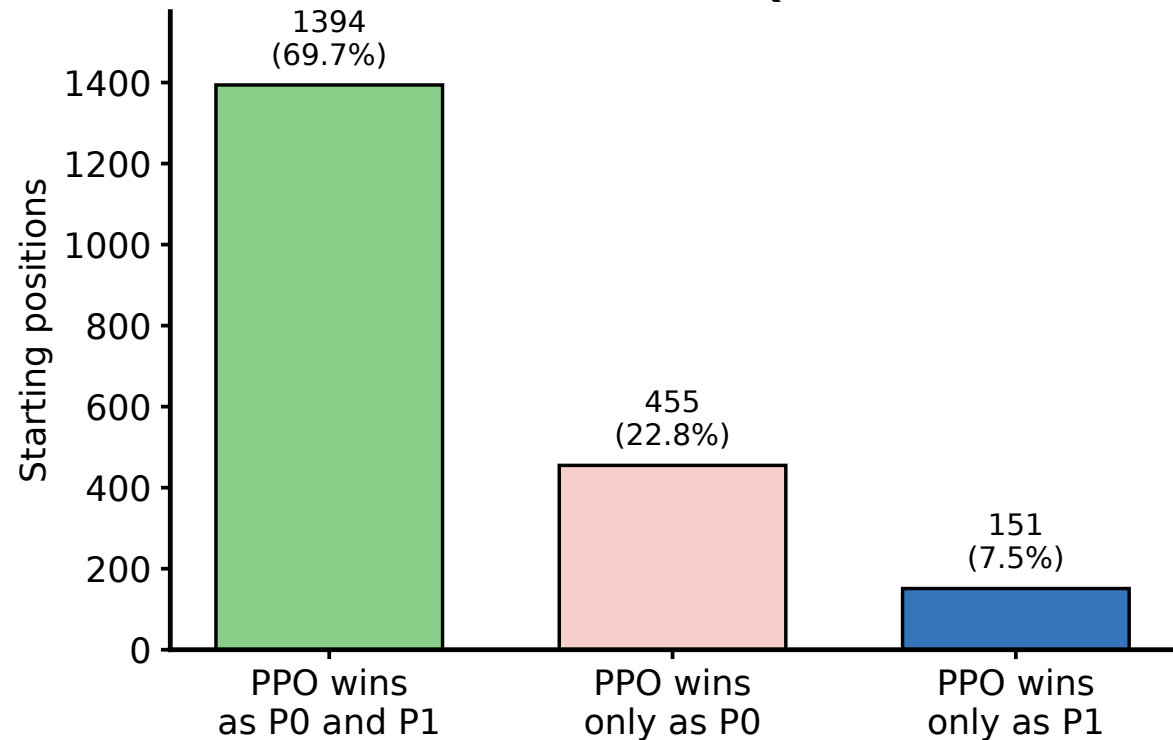
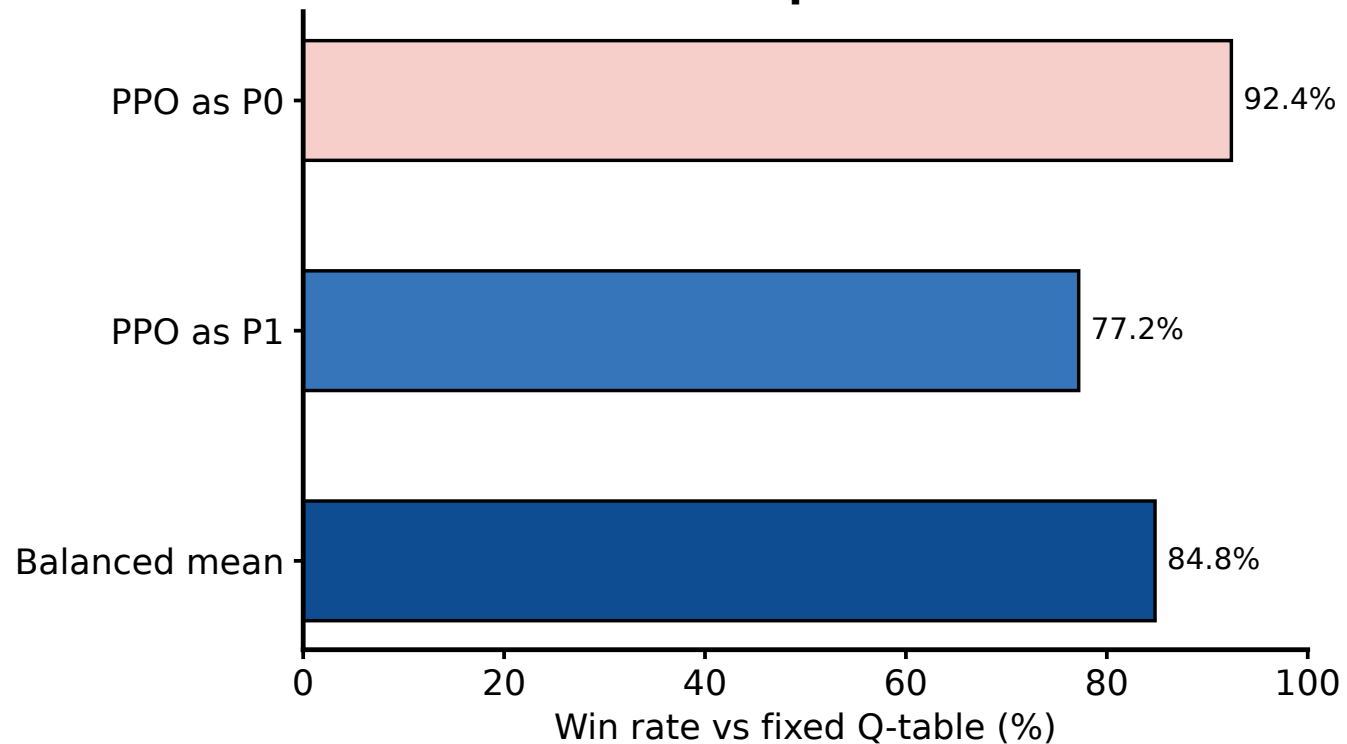


**a Deterministic PPO vs Q-table**



**b Role-conditioned performance**



n = 2,000 fixed starting idioms, two games per start. Head-to-head outcomes do not certify minimax values.